Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about radioactivity and its uses.

Which of these radiations is used in smoke detectors?

(1)

☐ **A** alpha
☐ **B** beta minus
☐ **C** beta plus
☐ **D** gamma

(Total for question = 2 marks)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about radioactivity and its uses.

Which of these radiations does **not** have a charge?

(1)

☐ **A** alpha
☐ **B** beta minus
☐ **C** beta plus
☐ **D** gamma

(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The nucleus of a different atom emits a gamma ray.

What happens to the number of particles in the nucleus?

(1)

☐ **A** it decreases by one
☐ **B** it decreases by two
☐ **C** it decreases by four
☐ **D** it does not change

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

(i) Which of these is the unit for the activity of a radioactive isotope?

(1)

☐ **A** second (s)
☐ **B** kilogram (kg)
☐ **C** kelvin (K)
☐ **D** becquerel (Bq)

(ii) The isotope nitrogen-13 (N-13) has a half-life of 10 minutes.

A sample of N-13 has an initial mass of 120 g.
Calculate the mass of N-13 that remains after 30 minutes.

(2)

mass remaining = ... g

(Total for question = 3 marks)

Q5.

The typical size of an atom is

(1)

☐ **A** 10^{-5} m

☐ **B** 10^{-10} m

☐ **C** 10^{-15} m

☐ **D** 10^{-20} m

(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

This question is about using radioactivity and nuclear energy.

(i) Gamma ray irradiation of food kills harmful bacteria by using

(1)

☐ **A** ionising particles
☐ **B** ionising radiation
☐ **C** non-ionising particles
☐ **D** non-ionising radiation

(ii) Which part of a nuclear reactor is used to absorb neutrons?

(1)

☐ **A** control rods
☐ **B** coolant
☐ **C** moderator
☐ **D** PET scanner

(Total for question = 2 marks)

Q7.

Answer the question with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 8 shows the symbol for the nucleus of an atom of strontium-90.

$^{90}_{38}Sr$

Figure 8

(i) How many protons are in the nucleus of an atom of strontium-90?

(1)

- ☐ **A** 38
- ☐ **B** 52
- ☐ **C** 90
- ☐ **D** 128

(ii) How many neutrons are in the nucleus of an atom of strontium-90?

(1)

- ☐ **A** 38
- ☐ **B** 52
- ☐ **C** 90
- ☐ **D** 128

(Total for question = 2 marks)

Q8.

This question is about radioactivity and its uses.

The diagram in Figure 13 shows a radioactive source used to check the thickness of paper.

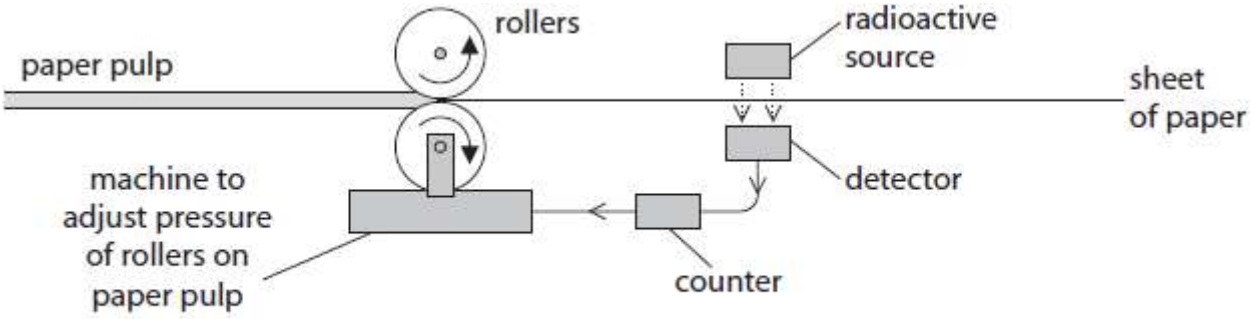

Figure 13

(i) Name the type of radiation used to check the thickness of the paper.

(1)

.....

(ii) Give the name of **one** device which could detect this type of radiation.

(1)

.....

When the paper is the correct thickness, the count rate is 4000 counts per minute.

(iii) The count rate increases when the paper gets thinner.
Give **one** reason for this increase.

(1)

.....

.....

(iv) The rollers need to be adjusted if the count rate increases by 5%.

Calculate the maximum count rate that would be allowed before the rollers need to be adjusted.

(2)

maximum count rate = ... counts per minute

(Total for question = 5 marks)

Q9.

Power stations that use nuclear reactors produce high level radioactive waste.

State **one** way that this radioactive waste can be disposed of safely.

(1)

.....

.....

(Total for question = 1 mark)

Q10.

Some isotopes are unstable.
They emit β^- particles when they decay.

Explain how a nucleus changes when a β^- particle is emitted.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Mark Scheme

Q1.

	Answer	Mark
	A alpha **B,C,** and **D** travel too far in air and are not sufficiently ionising	**(1)** **AO1**

Q2.

	Answer	Mark
	D gamma **A,B** and **C** all carry a charge	**(1)** **AO1**

Q3.

Question number	Answer	Additional guidance	Mark
	D it does not change A, B and C are incorrect because the number of nucleons does not change in gamma emission		**1** **AO1.1**

Q4.

Q5.

Question Number	Answer	Mark
	B 10^{-10} m	**(1)**

Q6.

Q7.

Question number	Answer	Mark
(i)	☒ **A** 38 B is number of neutrons C is mass number D is an irrelevant addition of two numbers	**(1)**

Question number	Answer	Mark
(ii)	☒ **B** 52 A is number of protons C is mass number D is an irrelevant addition of two numbers	**(1)**

Q8.

	Answer	Additional guidance	Mark
(i)	beta, beta (minus), β, β^-	accept beta(plus), β^+	**(1)** **AO1**

	Answer	Additional guidance	Mark
(ii)	Geiger-Muller (tube)/ G-M (tube)/ GM (tube)/ Geiger (counter)	accept ratemeter do not accept radiation meter accept phonetic spellings e.g. Giga	**(1)** **AO1**

	Answer	Additional guidance	Mark
(iii)	one from more (radiation) passes through (the paper) (1) OR less (radiation) is absorbed (by the paper) (1) OR less paper (for radiation) to pass through/penetrate(1)	penetrates/gets through allow easier to pass through Ignore name of particle if given	**(1)** **AO2**

	Answer	Additional guidance	Mark
(iv)	Calculation of 5% of count rate (1) $\frac{4000 \times 5}{100}$ or 4000 x 0.05 = 200 evaluation (1) 4000 +200 = 4200	accept use of any percentage between 4% and 5% accept any value between 160 and 200 ecf for addition of incorrect calculated percentage and 4000 award full marks for any answer that rounds to 4200 without working	**(2)** **AO2**

Q9.

Question number	Answer	Additional guidance	Mark
	encapsulated in glass and/or buried deep underground		**(1)**

Q10.

Question number	**Answer**	**Additional guidance**	**Mark**
	An explanation that combines understanding (1 mark) and reasoning (1 mark) linking: • number of neutrons decreases by one (1) • number of protons increases by one.(1)	a neutron becomes a proton plus an electron for (2) marks	**(2)**